

IMO Phase II Level 2 Training: Extremal Combinatorics

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23rd September, 2021

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1 Introduction

Definition 1.1 (Basic Definitions)

1. For a set X , $|X|$ denote its size.
2. For positive integer n , let $[n] = \{1, 2, \dots, n\}$, the power set $2^{[n]}$ consists of 2^n subsets (including $[n]$ and the empty set $\emptyset$) of $[n]$.
3. Let $\binom{[n]}{k}$ denote the collection of all k -element subsets of $[n]$.
4. A subset $\mathcal{F}$ of $2^{[n]}$ is called a **family** of subsets. (Think of $\mathcal{F} \subseteq 2^X$ as a hypergraph)
5. A family $\mathcal{F} \subseteq \binom{[n]}{k}$ is called k -uniform.

Remark. Sometimes a family $\mathcal{F} \subseteq 2^X$ is called **non-uniform**.

Notation 1.2 (Basic Notation)

1. $\mathcal{F}(x) = \{F \setminus \{x\} : x \in F \in \mathcal{F}\},$
2. $\mathcal{F}(\bar{x}) = \{F : x \notin F \in \mathcal{F}\}.$

Definition 1.3 (Chain)

We call $(F_0, \dots, F_l)$ a **chain** of length l if

$$F_0 \subsetneq F_1 \subsetneq \dots \subsetneq F_l.$$

If for a family $\mathcal{F}$ no $F, G \in \mathcal{F}$ satisfy $F \subsetneq G$, $\mathcal{F}$ is called an **antichain**.

Remark. $\binom{[n]}{k}$ is an antichain for every fixed $k, 0 \leq k \leq n$.

Theorem 1.4 (Sperner's Theorem)

Let $\mathcal{F} \subseteq 2^{[n]}$ be an antichain. Then

$$|\mathcal{F}| \leq \binom{n}{\lfloor \frac{n}{2} \rfloor},$$

and in the case of equality, $\mathcal{F} = \binom{[n]}{\frac{n}{2}}$ for n even and $\mathcal{F} = \binom{[n]}{\frac{n-1}{2}}$ or $\binom{[n]}{\frac{n+1}{2}}$ for n odd.

Proof. Let $\mathcal{F} \subseteq 2^{[n]}$ be an maximal antichain. Further let

$$p = \max \{|F| : F \in \mathcal{F}\} \text{ and } q = \min \{|F| : F \in \mathcal{F}\}.$$

If we can show $p \leq \frac{n+1}{2}$ and $q \geq \frac{n-1}{2}$, then the proof for n even is complete (it implies $p = q = \frac{n}{2}$). If n is odd, we have $\mathcal{F} \subseteq \binom{[n]}{\frac{n-1}{2}} \cup \binom{[n]}{\frac{n+1}{2}}$, which requires some more work.

We are going to proof by contradiction. Assume $p > \frac{n+1}{2}$, and define $\mathcal{G} = \{G \in \mathcal{F} : |G| = p\}$. We then define the immediate shadow $\mathcal{H}$ of $\mathcal{G}$ by

$$\mathcal{H} = \{H : |H| = p-1 \text{ and } H \subseteq G \text{ for some } G \in \mathcal{G}\}.$$

Claim 1.5

We have the following:

1. Let $\mathcal{F}' = (\mathcal{F} \setminus \mathcal{G}) \cup \mathcal{H}$. Then $\mathcal{F}'$ is an antichain.
2. $|\mathcal{F}'| > |\mathcal{F}|$.

With this claim, the $p \leq \frac{n+1}{2}$ is complete because $|\mathcal{F}'| > |\mathcal{F}|$ contradicts the maximal choice of antichain $\mathcal{F}$. Similar argument for proving $q \geq \frac{n-1}{2}$. Yet, we may circumvent by consider the complement $\mathcal{F}^c = \{[n] \setminus F : F \in \mathcal{F}\}$, then check $|\mathcal{F}^c| = |\mathcal{F}|$ and $\mathcal{F}^c$ is an antichain, too. Then we have $n - |F| \leq \frac{n+1}{2} \forall F \in \mathcal{F}$, $|F| \geq \frac{n-1}{2}$. $\square$

Proof. (claim 1.5) For (i), We have $\mathcal{H}$ an antichain (as $\mathcal{H} \subseteq \binom{[n]}{p-1}$) and $H \not\subseteq F$ for $H \in \mathcal{H}, F \in \mathcal{F} \setminus \mathcal{G}$ (as $\max \{|F| : F \in \mathcal{F} \setminus \mathcal{G}\} \leq p-1$). Since for $H \in \mathcal{H}$, exist $G \in \mathcal{G}$ with $H \subseteq G$, $H \supseteq F$ implies $G \supseteq F$, contradicts $\mathcal{F} = (\mathcal{F} \setminus \mathcal{G}) \cup \mathcal{G}$ is an antichain. Hence $H \not\supseteq F$.

For (ii), since $|\mathcal{F}'| - |\mathcal{F}| = |\mathcal{H}| - |\mathcal{G}|$. We will prove by double counting. Let M be the number of pairs (G, H) with $G \in \mathcal{G}, H \in \mathcal{H}$. Since every p -element subset G has p subsets of $p-1$, $M = p|\mathcal{G}|$. And since every $(p-1)$ -element $H \subseteq [n]$ there are $n - (p-1) = n+1-p$ choices of $G' \in \binom{[n]}{p}, H \subseteq G'$ (some of G' may lie outside of $\mathcal{G}$), we have

$$p|\mathcal{G}| = M \leq (n-p+1)|\mathcal{H}|.$$

If $p > \frac{n+1}{2}$, $n+1-p < \frac{n+1}{2}$, hence $|\mathcal{G}| < |\mathcal{H}|$. $\square$

Definition 1.6 (Intersecting)

A family $\mathcal{F} \subseteq 2^{[n]}$ is called **intersecting** if $F \cap F' \neq \emptyset$ for all $F, F' \in \mathcal{F}$.

Theorem 1.7 (Erdos-Ko-Rado Theorem)

Suppose that $\mathcal{F} \subseteq 2^{[n]}$ is an intersecting family. Then:

1. $|\mathcal{F}| \leq 2^{n-1}$.
2. If $\mathcal{F} \subseteq \binom{[n]}{k}$ and $n \geq 2k$, then $|\mathcal{F}| \leq \binom{n-1}{k-1}$.

Proof. For (i), let $A \subseteq [n]$, $A^c = [n] \setminus A$. If $F \in \mathcal{F}$, then $F \cap F^c = \emptyset$ implies $F^c \notin \mathcal{F}$. Hence $\mathcal{F} \cap \mathcal{F}^c = \emptyset$, where $\mathcal{F}^c = \{F^c : F \in \mathcal{F}\}$. Since $|\mathcal{F}| = |\mathcal{F}^c|$ and $\mathcal{F} \sqcup \mathcal{F}^c \subseteq 2^{[n]}$, we have

$$2|\mathcal{F}| = |\mathcal{F}| + |\mathcal{F}^c| = |\mathcal{F} \sqcup \mathcal{F}^c| \leq |2^{[n]}| = 2^n.$$

(ii) will be covered by the end of the lesson. □

2 Operations on Sets and Set Systems

Definition 2.1 (Set Operations)

For $A, B \subseteq X$, we have four natural operations:

$$A \cap B, A \cup B, A \setminus B = \{x \in A : x \notin B\}, A \triangle B = (A \setminus B) \cup (B \setminus A),$$

the last one is called **symmetric difference**.

We define **distance** $d(A, B) = |A \triangle B|$, 2^X becomes a **metric space**.

Remark. Check that $d(A, b) = 0$ iff $A = B$, and $d(A, C) \leq d(A, B) + d(B, C)$.

Definition 2.2 (Complement)

For $A \subseteq X$, the set $X \setminus A$ is called the **complement** of A , and if X is fixed we simply denote it as A^c . For a family $\mathcal{F} \subseteq 2^X$, the **complement family** is defined by $\mathcal{F}^c = \{F^c : F \in \mathcal{F}\}$.

Remark. We have $A \cap A^c = \emptyset$, $A \cup A^c = X$, and $|A \cup B| + |A \cap B| = |A| + |B|$.

2.1 Squash

Definition 2.3 (Squash)

Let $\mathcal{F} \subseteq 2^X$, $x \in X$ be a fixed element. The operation **squash** at an element x is defined by

$$S_x(\mathcal{F}) = \{F_x : F \in \mathcal{F}\},$$

where

$$F_x = \begin{cases} F \setminus \{x\} & \text{if } x \in F \subseteq \mathcal{F} \text{ and } F \setminus \{x\} \notin \mathcal{F}, \\ F & \text{otherwise.} \end{cases}$$

Remark. By definition, we have $|S_x(\mathcal{F})| = |\mathcal{F}|$.

Example 2.4

If $\mathcal{F} = \{\{x, y, z\}, \{x, y\}, \{y, z\}, \{z\}\}$, then $S_x(\mathcal{F}) = \{\{x, y, z\}, \{y\}, \{y, z\}, \{z\}\}$.

Observation 2.5

Let $F \in S_x(\mathcal{F})$ and $x \in F$. Then both F and $F \setminus \{x\}$ are members of $\mathcal{F} \cap S_x(\mathcal{F})$.

Definition 2.6 (Diameter)

We define **diameter** $\text{diam}(\mathcal{F})$ of $\mathcal{F}$ by

$$\text{diam}(\mathcal{F}) = \max_{A, B \in \mathcal{F}} |A \Delta B|.$$

Proposition 2.7

$$\text{diam}(S_x(\mathcal{F})) \leq \text{diam}(\mathcal{F}).$$

Proof. Consider the squash of $\mathcal{F}$ at x . Take arbitrary $A, B \in \mathcal{F}$, if $A_x \Delta B_x \subseteq A \Delta B$, then it is fine. The only other case is $A_x \Delta B_x = (A \Delta B) \sqcup \{x\}$, which we have $x \in A \cap B$. WLOG, let $x \in A_x, x \notin B_x$. By the observation, $A \setminus \{x\} \in \mathcal{F}$. Now $A_x \Delta B_x = (A \setminus \{x\}) \Delta B$ implies $\text{diam}(S_x(\mathcal{F})) \leq \text{diam}(S_x(\mathcal{F})) \leq \text{diam}(\mathcal{F})$. $\square$

Remark. Applying the operation squash at different elements $x, y \in X$ will produce a family independent of the order of operations, that is, $S_x(S_y(\mathcal{F})) = S_y(S_x(\mathcal{F}))$.

Definition 2.8 (Squashed family)

We denote $\tilde{\mathcal{F}}$ be the family after applying operation squash at all $x \in X$ at $\mathcal{F}$ just once. It has the property:
For all $x \in X$, if $x \in F \subseteq \tilde{\mathcal{F}}$ then $F \setminus \{x\} \in \tilde{\mathcal{F}}$.

Definition 2.9 (Hereditary)

A family $\mathcal{F}$ is **hereditary** (or a downset) if $G \subseteq F \in \mathcal{F}$ implies $G \in \mathcal{F}$.
In other words, $\mathcal{F}$ is hereditary if it is a union of power sets.

Proposition 2.10

A squashed family $\tilde{\mathcal{F}}$ is hereditary.

Proof. Let $G \subseteq F \in \mathcal{F}$. We show that $G \in \tilde{\mathcal{F}}$ by induction on $|F \setminus G|$. For $|F \setminus G| = 1$, it's just the property. For induction, choose H with $G \subsetneq H \subsetneq F$, then $F \in \mathcal{F}$ and induction yields $H \in \mathcal{F}$. And apply induction to (G, H) yields $G \in \mathcal{F}$. $\square$

Definition 2.11 (Trace)

For a family $\mathcal{F} \subseteq 2^X$ and a subset $T \subseteq X$, we define **trace** $\mathcal{F}_{|T}$ by

$$\mathcal{F}_{|T} = \{F \cap T : F \in \mathcal{F}\}.$$

Remark. $\mathcal{F}_{|T}$ is a usual family, i.e. subset of 2^T . Consequently, $|\mathcal{F}_{|T}| \leq 2^{|T|}$.

Proposition 2.12

Let $\mathcal{F} \subseteq 2^X$ and $x \in X$. For every $T \subseteq X$ one has

$$|S_x(\mathcal{F})_{|T}| \leq |\mathcal{F}_{|T}|.$$

Proof. If $x \notin T$, squash does not affect the trace on T . Let $T = \{x\} \sqcup T'$ and $T_0 \subseteq T'$. If $\{x\} \cup T_0 \in S_x(\mathcal{F})$, then $\{T_0, \{x\} \cup T_0\} \subseteq S_x(\mathcal{F}) \cap \mathcal{F}$ by the observation. If $\{x\} \cup T_0 \notin S_x(\mathcal{F})$ and $T_0 \in S_x(\mathcal{F})$, then $T_0 \in \mathcal{F}$ or

$\{x\} \cup T_0 \in \mathcal{F}$ by squash. Hence we always have

$$|S_x(\mathcal{F})|_T \cap \{T_0, \{x\} \cup T_0\}| \leq |\mathcal{F}|_T \cap \{T_0, \{x\} \cup T_0\}|.$$

Summing this for all $2^{|T|-1}$ choices of $T_0 \subseteq T'$, hence the equation. $\square$

2.2 Shift

Definition 2.13 (Shifting)

For $1 \leq i \neq j \leq n$, we are going to define the (i, j) -**shift** $S_{i,j}$ form families $\mathcal{F} \subseteq 2^{[n]}$. The formula is as follow:

$$S_{i,j}(\mathcal{F}) = \{s_{i,j}(F) : F \in \mathcal{F}\},$$

where

$$s_{i,j}(F) = \begin{cases} (F \setminus \{j\}) \cup \{i\} & \text{if } i \notin F, j \in F \text{ and } (F \setminus \{j\}) \cup \{i\} \notin \mathcal{F}, \\ F & \text{otherwise.} \end{cases}$$

Definition 2.14 (Intersecting and Shadow)

Let $t \in \mathbb{Z}^+$, we say a family $\mathcal{F} \subseteq 2^X$ is **t -intersecting** if $|F \cap F'| \geq t \forall F, F' \in \mathcal{F}$.

We simply say a family is **intersecting** if it is 1-intersecting.

For $p \in \mathbb{Z}^+, 1 \leq p \leq n$, define the **p -shadow** $\sigma_p(\mathcal{F})$ by

$$\sigma_p(\mathcal{F}) = \left\{ P \in \binom{[n]}{p} : \exists F \in \mathcal{F}, P \subseteq F \right\}.$$

Proposition 2.15

If $\mathcal{F} \subseteq 2^X$ is t -intersecting, then $S_{i,j}(\mathcal{F})$ is t -intersecting as well.

Proof. Suppose contradiction that exist $F, G \in \mathcal{F}$ with $|S_{i,j}(F) \cap S_{i,j}(G)| < t \leq |F \cap G|$. WLOG, assume $s_{i,j}(F) = F$ but $s_{i,j}(G) \neq G$, with $F \cap \{i, j\} = G \cap \{i, j\} = \{j\}$. The fact that $s_{i,j}(F) = F$ is because of that $F' := (F \setminus \{j\}) \cup \{i\} \in \mathcal{F}$. However, $|F' \cap G| = |s_{i,j}(F) \cap s_{i,j}(G)|$, contradicting the t -intersecting property of $\mathcal{F}$. $\square$

Proposition 2.16

$$\sigma_p(S_{i,j}(\mathcal{F})) \subseteq S_{i,j}(\sigma_p(\mathcal{F})).$$

Proof. Let $P \in \sigma_p(S_{i,j}(\mathcal{F}))$. If $|P \cap \{i, j\}| = 0$ or 2 , then we can find $F \in \mathcal{F}$ s.t. $P \subseteq F$, and it follows that $P \in \sigma_p(\mathcal{F})$ and $P \in S_{i,j}(\sigma_p(\mathcal{F}))$. So we may assume $|P \cap \{i, j\}| = 1$. Set $R = P \setminus \{i, j\}$. Since $P \in \sigma_p(S_{i,j}(\mathcal{F}))$, there is $Q \in S_{i,j}(\mathcal{F})$ s.t. $P \subseteq Q$.

First suppose $P = R \sqcup \{i\}$. If $Q \in \mathcal{F}$ then $P \in \sigma_p(\mathcal{F})$ and $P \in S_{i,j}(\sigma_p(\mathcal{F}))$. If $Q \notin \mathcal{F}$ then $F := (Q \setminus \{i\}) \cup \{j\} \in \mathcal{F}$ and $R \sqcup \{j\} \subseteq F$. Thus $R \sqcup \{j\} \in \sigma_p(\mathcal{F})$ and $P = R \sqcup \{i\} \in S_{i,j}(\sigma_p(\mathcal{F}))$.

Next suppose $P = R \sqcup \{j\}$. Then $Q \in \mathcal{F}$ and $P \in \sigma_p(\mathcal{F})$. If $i \in Q$ then $R \sqcup \{i\} \in \sigma_p(\mathcal{F})$. If $i \notin Q$ then $(Q \setminus \{j\}) \cup \{i\} \in \mathcal{F}$ and $R \sqcup \{i\} \in \sigma_p(\mathcal{F})$ again. Thus $s_{i,j}(P) = P$ and $P \in S_{i,j}(\sigma_p(\mathcal{F}))$. $\square$

Definition 2.17 (s-union)

We say that a family $\mathcal{G} \subseteq 2^X$ is **s -union** if $|G \cup G'| \leq |X| - s \forall G, G' \in \mathcal{G}$.

We say a family is **union** if it is 1-union.

Exercise 2.18

Let $s \in \mathbb{Z}^+$, $s \leq |X|$.

1. $\mathcal{G} \subseteq 2^X$ is s iff $\mathcal{G}$ is s -intersecting.
2. Show that if $\mathcal{G} \subseteq 2^X$ is s -union then $S_{i,j}(\mathcal{G})$ is s -union.

Exercise 2.19

Show that if a family $\mathcal{G} \subseteq 2^X$ is union, then $|\mathcal{G}| \leq 2^{|X|-1}$.

Remark. If $G := s_{i,j}(F) \neq F$ then $\sum_{a \in F} a - \sum_{a \in G} a = j - i$.

Observation 2.20

If $G := s_{i,j}(F) \neq F$ then $\sum_{a \in F} a - \sum_{a \in G} a = j - 1$. Hence if $1 \leq i < j \leq n$ and $S_{i,j}(\mathcal{F}) \neq \mathcal{F} \subseteq 2^X$, then

$$\sum_{F \in \mathcal{F}} \sum_{a \in F} a > \sum_{G \in S_{i,j}(\mathcal{F})} \sum_{a \in G} a.$$

Remark. Since the sum has a lower bound, after repeatedly apply (i, j) -shift for various $1 \leq i < j \leq n$, we can get the following:

Definition 2.21 (Shifted family)

After repeatedly applying (i, j) -shift, we end up with a family $\tilde{\mathcal{F}}$ satisfying $S_{i,j}(\tilde{\mathcal{F}}) = \tilde{\mathcal{F}} \forall 1 \leq i < j \leq n$. Such family is called **shifted**.

Remark. From proposition 2.15, it is clear that in establishing upper bounds for the size of t -intersecting (or s -union) families we can restrict ourselves to shifted families. And by proposition 2.16, the same is true when trying to prove lower bounds on the size of p -shadow.

Theorem 2.22 (Kruskal-Katona Theorem (integer version))

Let $0 \leq p < k \leq x$ be integers. Suppose that $\mathcal{F}$ is a family of k -element subsets with $|\mathcal{F}| \geq \binom{x}{k}$. Then $|\sigma_p(\mathcal{F})| \geq \binom{x}{p}$.

Proof. Let $p < k$ be fixed. We will prove by induction on x . $x = k$ is trivial. For $x \geq k + 1$, i.e. $x - 1 \geq k$. By proposition 2.16, we may suppose $\mathcal{F}$ is shifted. First consider $p = k - 1$.

Claim 2.23

$$\sigma_{k-1}(\mathcal{F}(\bar{1})) \subseteq \mathcal{F}(1).$$

Proof. Let $G \in \sigma_{k-1}(\mathcal{F}(\bar{1}))$. Then $G \cup \{j\} \in \mathcal{F}$ for some $j \notin \{1\} \sqcup G$. By shiftedness, $G \cup \{1\} \in \mathcal{F}$, i.e. $G \in \mathcal{F}(1)$. $\square$

Claim 2.24

$$|\mathcal{F}(1)| \geq \binom{x-1}{k-1}.$$

Proof. Suppose contrary, i.e. $|\mathcal{F}(1)| < \binom{x-1}{k-1}$. We infer that

$$|\mathcal{F}(\bar{1})| = |\mathcal{F}| - |\mathcal{F}(1)| > \binom{x}{k} - \binom{x-1}{k-1} = \binom{x-1}{k}.$$

Hence may apply induction to $\mathcal{F}(\bar{1})$, and get $|\sigma_{k-1}(\mathcal{F}(\bar{1}))| \geq \binom{x-1}{k-1} > |\mathcal{F}(1)|$. $\square$

Since $\sigma_{k-1}(\mathcal{F}) \supseteq \mathcal{F}(1) \sqcup \{\{1\} \sqcup G : G \in \sigma_{k-2}(\mathcal{F}(1))\}$. Clearly $\sigma_{k-1}(\mathcal{F})$ contains $\mathcal{F}(1)$, where $1 \notin \mathcal{F}(1)$. If $G \in \sigma_{k-2}(\mathcal{F}(1))$, then there is $H \in \mathcal{F}(1)$ s.t. $G \subseteq H$, and $\{1\} \sqcup G \subseteq \{1\} \sqcup H \in \mathcal{F}$, yielding that $\{1\} \sqcup G \in \sigma_{k-1}(\mathcal{F})$. Thus

$$|\sigma_{k-1}(\mathcal{F})| \geq |\mathcal{F}(1)| + |\sigma_{k-2}(\mathcal{F}(1))|.$$

Applying induction hypothesis by claim(2) to $\mathcal{F}(1)$, we infer $|\sigma_{k-2}(\mathcal{F}(1))| \geq \binom{x-1}{k-2}$. This proves

$$|\sigma_{k-1}(\mathcal{F})| \geq \binom{x-1}{k-1} + \binom{x-1}{k-2} = \binom{x}{k-1}.$$

We prove the general case $p < k$ by induction on $i := k - p$, where we proved $i = 1$. Suppose the statement is true for the case i , i.e., $|\mathcal{F}| \geq \binom{x}{k}$ implies $|\sigma_{k-i}(\mathcal{F})| \geq \binom{x}{k-i}$. Since $\sigma_{k-(i+1)}(\mathcal{F}) = \sigma_{(k-i)-1}(\sigma_{k-i}(\mathcal{F}))$, we can imply the induction hypothesis to the case $k' = k - i$ and $p' = k' - 1$ (i.e. $i' = 1$); we get $|\sigma_{k-(i+1)}(\mathcal{F})| = |\sigma_{k'-1}(\sigma_{k'}(\mathcal{F}))| \geq \binom{x}{k'-1} = \binom{x}{k-(i+1)}$. $\square$

3 Theorems on Traces

Theorem 3.1

Let $n > k \geq 0$ be integers. If $\mathcal{F} \subseteq 2^{[n]}$ satisfies $|\mathcal{F}| > \sum_{i=0}^k \binom{n}{i}$, then there exists a $T \in \binom{[n]}{k+1}$ such that $\mathcal{F}|_T = 2^T$, where trace $\mathcal{F}|_T = \{F \cap T : F \in \mathcal{F}\}$.

Proof. Suppose contrary, that $\mathcal{F} \subseteq 2^{[n]}$ with $\mathcal{F}|_T \neq 2^T$ for all $T \in \binom{[n]}{k+1}$. By proposition 2.10 and proposition 2.12 we can obtain a hereditary family $\mathcal{G} \subseteq 2^{[n]}$ with $|\mathcal{G}| = |\mathcal{F}|$, and still satisfying $|\mathcal{G}|_T| < 2^{|T|} \forall T \in \binom{[n]}{k+1}$. We claim that $|G| \leq k \forall G \in \mathcal{G}$.

Indeed, if $|G| \geq k+1$ for some $G \in \mathcal{G}$, then by hereditary property $2^G \subseteq \mathcal{G}$, and thus for all $H \in \binom{G}{k+1}$ we have $\mathcal{G}|_H = 2^H$, a contradiction. Consequently, we have proved that $|G| \leq k \forall G \in \mathcal{G}$ and $|\mathcal{G}| \leq \sum_{i=0}^k \binom{n}{i}$. But this contradict $|\mathcal{G}| = |\mathcal{F}| > \sum_{i=0}^k \binom{n}{i}$. $\square$

Definition 3.2 (Arrow Notation)

We define the **arrow notation** $(n, m) \rightarrow (a, b)$ with the following meaning:
For every $\mathcal{F} \subseteq 2^{[n]}$ with $|\mathcal{F}| = m$, there exists $T \subseteq [n]$ with $|T| = a$ such that $|\mathcal{F}|_T| \geq b$.

Theorem 3.3

The following hold.

1. $(n, m) \rightarrow (n-1, m)$ for $m \leq n$.
2. $(n, m) \rightarrow (n-1, m-1)$ for $m \leq 1 + n + \frac{n-1}{2}$.
3. $(n, m) \rightarrow (3, 7)$ for $m > 1 + n + \frac{n^2}{4}$.

Remark. By proposition 2.12, we just need to check for hereditary families.

Proof. 1. If $\mathcal{G} \subseteq 2^{[n]}$ hereditary, then $\emptyset \in \mathcal{G}$. If, moreover, $m = |\mathcal{G}| \leq n$, then exist $x \in [n]$ s.t. $\{x\} \notin \mathcal{G}$. Hence $G \notin \mathcal{G} \forall G$ where $x \in G$. Hence $\mathcal{G} \subseteq 2^{[n] \setminus \{x\}}$, $|\mathcal{G}|_{[n] \setminus \{x\}}| = |\mathcal{G}|$.

2. If $\{x\} \notin \mathcal{G}$ for some $m \in [n]$, the above works. Assume $\emptyset \in \mathcal{G}, \{x\} \in \mathcal{G} \forall x$. Since $m = |\mathcal{G}| \leq 1 + n + \frac{n-2}{2}$, there is at most $\lfloor \frac{n-1}{2} \rfloor$ more subsets in $\mathcal{G}$. Hence exist $y \in [n]$ not contained in any 2-element member of $\mathcal{G}$. By the hereditary property, the only member of $\mathcal{G}$ containing y is $\{y\}$. Hence $\mathcal{G}|_{[n] \setminus \{y\}} = \mathcal{G} \setminus \{y\}$.

3. If $|G| = 3$ for some $G \in \mathcal{G}$, then $|\mathcal{G}|_G| = 8$ and the result follows. Assume $\mathcal{G} \subseteq \binom{[n]}{0} \sqcup \binom{[n]}{1} \sqcup \binom{[n]}{2}$. Let $\mathcal{G}^{(i)} = \mathcal{G} \cap \binom{[n]}{i}$, then $\mathcal{G}^{(2)}$ is a graph on n vertices. The number of edges is at least $m - |\mathcal{G}^{(0)}| - |\mathcal{G}^{(1)}| \geq m - 1 - n > \frac{n^2}{4}$ by the assumption. We can then prove that exist edges $\{x, y\}, \{x, z\}$ and $\{y, z\}$ (forming a triangle). Now let $T = \{x, y, z\}$ and get $\mathcal{G}|_T = \binom{T}{0} \sqcup \binom{T}{1} \sqcup \binom{T}{2}$, and hence $|\mathcal{G}|_T| = 1 + 3 + 3 = 7$.

□

Exercise 3.4

Let G be a graph on n vertices with more than $n^2/4$ edges. Show that G contains a triangle.

Corollary 3.5

For any $\mathcal{F} \subseteq 2^X$ and $Y \subseteq X$, the arrow relation

$$(|X|, |\mathcal{F}|) \rightarrow (|Y|, |\mathcal{F}_Y| + 1)$$

is not true.

Definition 3.6 (Hereditary family)

Let n be of the form $n = dq$, $d, q \in \mathbb{Z}^+$. Let $X = X_1 \sqcup \dots \sqcup X_q$ be a partition with $|X_i| = d$, $1 \leq i \leq q$. Define a **hereditary family** $\mathcal{F} := \mathcal{F}(d, q) = 2^{X_1} \cup \dots \cup 2^{X_q}$. Note that $2^{X_i} \cap 2^{X_j} = \{\emptyset\}$ and $|\mathcal{F}| = 1 + (2^d - 1)\frac{n}{d}$. If $Y = X \setminus \{x\}$ for some $x \in X$, then $\mathcal{F}_Y = \mathcal{F} \setminus \{F \in \mathcal{F} : x \in F\}$. So it follows that $|\mathcal{F}_Y| = |\mathcal{F}| - 2^{d-1}$ for all $Y \subseteq \binom{X}{n-1}$. Thus the following statement is not true.

$$(dq, 1 + (2^d - 1)\frac{n}{d}) \rightarrow (dq - 1, 1 + (2^d - 1)\frac{n}{d} - 2^{d-1} + 1)$$

Remark. $\mathcal{F}(d, q)$ is an extremal example by the following theorem.

Theorem 3.7

Let $n, d \in \mathbb{Z}^+$, $n > d$ being fixed. Let $|X| = n$ and let $\mathcal{F} \subseteq 2^X$ be a hereditary family satisfying $|\mathcal{F}| \leq 1 + \frac{2^d - 1}{d}n$. Then one of the following holds:

1. $|\mathcal{F}(x)| < 2^{d-1}$ for some $x \in X$. In this case $|\mathcal{F}_Y| > |\mathcal{F}| - 2^{d-1}$ for $Y = X \setminus \{x\}$.
2. $\mathcal{F}$ is isomorphic to $\mathcal{F}(d, q)$ for some q (in particular, d divides n). In this case $|\mathcal{F}_Y| = |\mathcal{F}| - 2^{d-1}$ for all $Y \in \binom{X}{n-1}$.

Open Problem 3.8

For each n and s , determine or estimate the maximum value $m = m(n, s)$ such that $(n, m) \rightarrow (n-1, m-s)$.

Remark. We have $m(n, 0) = 0$ and $m(n, 1) = \lfloor (3n+1)/2 \rfloor$. By previous theorem, $m(n, 2^{d-1}) \geq 1 + (2^d - 1)\frac{n}{d}$. It can be proved that $m(n, 2) = 2n$. However, $m(n, s)$ is unknown in general.

Exercise 3.9

Show that $m(n, 2) = 2n$.

4 Erdos-Ko-Rado Theorem

Theorem 4.1 (Erdos-Ko-Rado Theorem)

Let $n, k \in \mathbb{Z}^+$, $n \geq 2k$. Suppose that $\mathcal{F} \subseteq \binom{[n]}{k}$ is intersecting, that is $F \cap F' \neq \emptyset \forall F, F' \in \mathcal{F}$. Then

$$|\mathcal{F}| \leq \binom{n-1}{k-1}.$$

Proof. We apply double induction on n and k . The statement is clearly true for $k = 1$ and all $n \geq 2$. The other base case is $n = 2k$: If $n = 2k$ then $\binom{2k}{k} = 2\binom{2k-1}{k-1}$. Then statement is equivalent to $2|\mathcal{F}| \leq \binom{2k}{k}$. Consider $\mathcal{F}^c = \{[n] \setminus F : F \in \mathcal{F}\}$. Then $\mathcal{F}^c \subseteq \binom{[2k]}{k}$, $|\mathcal{F}^c| = |\mathcal{F}|$ and $\mathcal{F} \cap \mathcal{F}^c = \emptyset$ since $\mathcal{F}$ is intersecting. Thus $\mathcal{F} \sqcup \mathcal{F}^c \subseteq \binom{[2k]}{k}$ and $|\mathcal{F} \sqcup \mathcal{F}^c| = 2|\mathcal{F}|$, $2|\mathcal{F}| \leq \binom{2k}{k}$.

Now let $k > 1$ be fixed. Suppose the statement holds for all (n', k') with $n' \geq 2k'$ and $k' < k$ and for all (n', k) with $n > n' \geq 2k$. By proposition 2.15, we may assume $\mathcal{F}$ is shifted.

Lemma 4.2

Both $\mathcal{F}(\bar{n})$ and $\mathcal{F}(n)$ are intersecting.

Proof. Since $\mathcal{F}(\bar{n}) \subseteq \mathcal{F}$, $\mathcal{F}(\bar{n})$ is intersecting. Assume contrary that exist $G, H \in \mathcal{F}(n)$ with $G \cap H = \emptyset$. Since $|G| + |H| = 2(k-1) < n-2$, there exists $i \in [n-1]$ s.t. $i \notin G \cup H$. As $\mathcal{F}$ is shifted, the set $H \cup \{i\}$ is also in $\mathcal{F}$. However,

$$(H \cup \{i\}) \cap (G \cup \{n\}) = H \cap G = \emptyset,$$

contradiction arises. □

Applying induction hypothesis to both $\mathcal{F}(\bar{n}) \subseteq \binom{[n-1]}{k}$ and $\mathcal{F}(n) \subseteq \binom{[n-1]}{k-1}$ gives us

$$|\mathcal{F}| = |\mathcal{F}(\bar{n})| + |\mathcal{F}(n)| \leq \binom{(n-1)-1}{k-1} + \binom{(n-1)-1}{(k-1)-1} = \binom{n-1}{k-1}.$$

□

5 Problems for Discussion

Problem 1. A subset M of $\{1, 2, \dots, 2006\}$ has the property that for any three elements x, y, z of M with $x < y < z$, $x + y$ does not divide z . Determine the largest possible size of M .

Ans. Consider the stronger condition that there does not exist distinct elements $x, y, z \in M$ such that $x + y = z$

Suppose $M = \{a_1, a_2, \dots, a_k\}$ is a subset of such elements. In the usual way, let $M' = \{a_2 - a_1, a_3 - a_1, \dots, a_k - a_1\}$. We must have $M \setminus \{a_1\}$ and M' disjoint, else we would have $a_i - a_1 = a_j \Leftrightarrow a_i = a_j + a_1$, contradiction. Hence $|M \setminus \{a_1\}| + |M'| = |M \setminus \{a_1\} \cup M'| = 2(k-1) \leq 2006$ hence $|M| = k \leq 1004$. The set $M = \{1003, 1004, \dots, 2006\}$ is a set with 1004 elements that satisfies the condition so we are done.

Problem 2. Let S be a subset of $\{1, 2, \dots, 2019\}$ with the following property: for any three numbers $a, b, c \in S$ we have $2020 \nmid a + b + c$ and $2020 \nmid a + b - c$. Determine the largest possible number of elements that S can have.

Ans.

We are going to prove that $|S| \leq 1010$

Let c be the smallest element of the S then consider the sets: $(c+1, 2019), (c+2, 2018), \dots, (c+k, 2020-k)$ Such that k is the max number from which $c+k \leq 2020-k$

Now we can see that if we have two elements for one of these k subject then we can take this elements and the c and we have: $a + b = 2020 + c$ which means $2020 \mid a + b - c = 2020$ So from each subject we can take at most 1 element

Therefore $\max |S| = k + 1$ but also we know that $c + 2k \leq 2020$ so $k \leq 1009$ which gives $\max |S| \leq 1010$

The number 1010 can be achieved we the set $S = \{1, 3, 5, \dots, 2019\}$

Problem 3. (Korean Summer Program Practice Test 2016 5) Find the maximal possible n , where $A_1, \dots, A_n \subseteq \{1, 2, \dots, 2016\}$ satisfy the following properties.

1. For each $1 \leq i \leq n$, $|A_i| = 4$.

2. For each $1 \leq i < j \leq n$, $|A_i \cap A_j|$ is even.

Ans. I think the above is poorly done, but on the right track. If I am not mistaken, the real answer is $\binom{1008}{2}$. The construction is simple: partition $\{1, 2, \dots, 2016\}$ into 1008 sets of size two, and take the family of all pairwise unions.

Denote the family of sets under consideration as $\mathcal{A} = \{A_1, A_2, \dots\}$. Again, we observe that if as j spans the set of non- i indices, $A_i \cap A_j$ has at most two disjoint nonempty outputs, all of whom have size 2. With that end in mind, if $A_i \cap A_j = S$ where $i \neq j$ and $S \neq \emptyset$, then we assign the elements of S a shared color distinct from any previously assigned colors, and carry out this process over all distinct i, j . If by the end m colors are assigned, then $|\mathcal{A}| \leq \binom{m}{2} + \frac{2016-2m}{4}$. Furthermore, we have $m \leq 1008$, and it follows that the maximum occurs at $m = 1008$, who gives the aforementioned construction $\square$

Problem 4. Let X be a set with n elements. Find the largest number of subsets of X , each with 3 elements, so that no two of them are disjoint.

Ans. Particular case of the famous Erdős-Ko-Rado theorem for intersecting families of equicardinal sets. For $n \geq 6$, the answer is $\binom{n-1}{2}$; for $n < 6$ simple casework.

Problem 5. (Roumanian JBMO team selection test 2013) A set that contains 7 elements is given. All its subsets containing 3 elements are colored with n colors such that every two disjoint subsets have different colors. Find the least value of n for which such a coloring is possible.

Ans. Let's color first three element subsets A of $\{1, 2, 3, 4, 5, 6\}$ with red and blue:

If $1 \in A$ then A is red, otherwise blue.

Now color all other three element subset of $\{1, 2, 3, 4, 5, 6, 7\}$ with green (they all have 7 in it).

So $n \leq 3$. Suppose $n = 2$.

Since for each pair A, B of subsets in monocolour family satisfy $|A \cap B| \geq 1$ it follow from Erdos-Ko-Rado Theorem that cardinality of this family is no more then $\binom{7-1}{3-1} = 15$. But then we can color at most 30 subsets. Contradiction, since there is $\binom{7}{3} = 35$ three element subsets in $\{1, 2, 3, 4, 5, 6, 7\}$.

Problem 6. (IRAN national math olympiad(3rd round)-2010-combinatorics-p1) Suppose that $\mathcal{F} \subseteq \binom{[n]}{k}$. We know that for every three distinct elements of $\mathcal{F}$ like A, B, C , at most one of $A \cap B, B \cap C$ and $C \cap A$ is $\emptyset$. for $k \leq \frac{n}{2}$ prove that:

$$|\mathcal{F}| \leq \max(1, 4 - \frac{n}{k}) \times \binom{n-1}{k-1}.$$

Ans. We can split the question into the following cases:

1. All the intersections are nonempty - then by Erdos-Ko-Rado $|F| \leq \binom{n-1}{k-1}$.
2. $\exists i, j$ s.t $A_i \cap A_j = \emptyset$. Notice that, then $\forall t \neq i, j$ we have $A_t \cap A_i \neq \emptyset$ and $A_t \cap A_j \neq \emptyset$.

Now, consider random permutation π of set X and arrange $\pi(1), \dots, \pi(n)$ on circle by clockwise order. Consider some random set P generated by random i s.t $P = \{\pi(i), \dots, \pi(k+i-1)\}$.

Let's count now the number of elements of F on circle such that they contain k consecutive numbers (label it by u and call such sets good).

- (a) If on the circle there are no good sets with empty intersection, we have $u \leq k$. So if we let $\mathbb{P}$ be the probability that $P \in F$, then

$$\mathbb{P} = \frac{|F|}{\binom{n}{k}} \leq \frac{k}{n}$$

so it follows that $|F| \leq \binom{n-1}{k-1} \cdot (*)$

- (b) If there are good sets with empty intersections, let them be A_i, A_j . By previous, any A_t must have non-empty intersection with both of them. Let the one distance between endpoints of A_i, A_j be a and let the other one be b . Clearly, $a + b = n - 2k$ i.e. $(k - a) + (k - b) = 4k - n$.

On the other hand, we must have $u \leq (k - a - 1) + (k - b - 1) + 2 = 2k - (a + b) = 4k - n$. Hence, as in 2.1 we have

$$\mathbb{P} = \frac{|F|}{\binom{n}{k}} \leq \frac{4k - n}{n} = \frac{k}{n} \cdot \left(4 - \frac{n}{k}\right) \implies |F| \leq \left(4 - \frac{n}{k}\right) \cdot \binom{n-1}{k-1} (**)$$

So, from $(*)$, $(**)$ result follows.

Problem 7. Let S be a set with n elements, and F be a family of subsets of S with 2^{n-1} elements, such that for each $A, B, C \in F$, $A \cap B \cap C$ is not empty. Prove that the intersection of all of the elements of F is not empty.

Ans. Let $\mathcal{F}_n$ be the family of all subsets of $[n]$. Let $\mathcal{F}_n(k)$ be the subfamily of elements of $\mathcal{F}$ that contain $k \in [n]$. Let the family in the problem be $\mathcal{G}$. Observe the natural poset isomorphism $\mathcal{F}_n(k) \cong \mathcal{F}_{n-1}$. Let P_n be the partition of $\mathcal{F}_n$ into pairs of complements. Suppose that the claim is false; that for every $k \in [n]$, there is a $S \in \mathcal{G}$ so that $k \notin S$. Clearly $\mathcal{G}$ has at most one element from each pair in P_n , and since there are 2^{n-1} pairs in P_n , $\mathcal{G}$ must have exactly one element from each pair. Consider the partition of $\mathcal{F}_n(k)$ induced by the natural poset isomorphism above and P_{n-1} (call it $P_n(k) \cong P_{n-1}$). Note that each pair in $P_n(k)$ intersects to form the singleton $\{k\}$. But no two elements of $\mathcal{G}$ can intersect in a singleton; otherwise we can find by hypothesis some element of $\mathcal{G}$ not containing said singleton, and the intersection of these three sets is empty, a contradiction. So there are at most 2^{n-2} elements of $\mathcal{G}$ in any $\mathcal{F}_n(k)$. It follows that $\mathbb{P}_{S \in \mathcal{G}}(k \in S) \leq 0.5$, and by Linearity of expectation we see that $\mathbb{E}_{S \in \mathcal{G}}(|S|) \leq 0.5n$. Now partition $\mathcal{F}_n$ into families $K_0, K_1, \dots, K_n$ where $S \in K_i \iff |S| = i$. Note that if $S \in K_i$ then the element of $\mathcal{F}_n$ paired S to by P_n is in K_{n-i} . Finally, $\mathcal{G} \cap K_i$ must respect the Erdős-Ko-Rado Theorem when $i \leq 0.5n$, since there are no two sets in $\mathcal{G}$ with empty intersection. But $i < 0.5n \implies \frac{\binom{n-1}{i-1}}{\binom{n}{i}} = \frac{i}{n} < 0.5$, so $\mathbb{E}_{S \in \mathcal{G}}(|S|) > 0.5n$, a contradiction